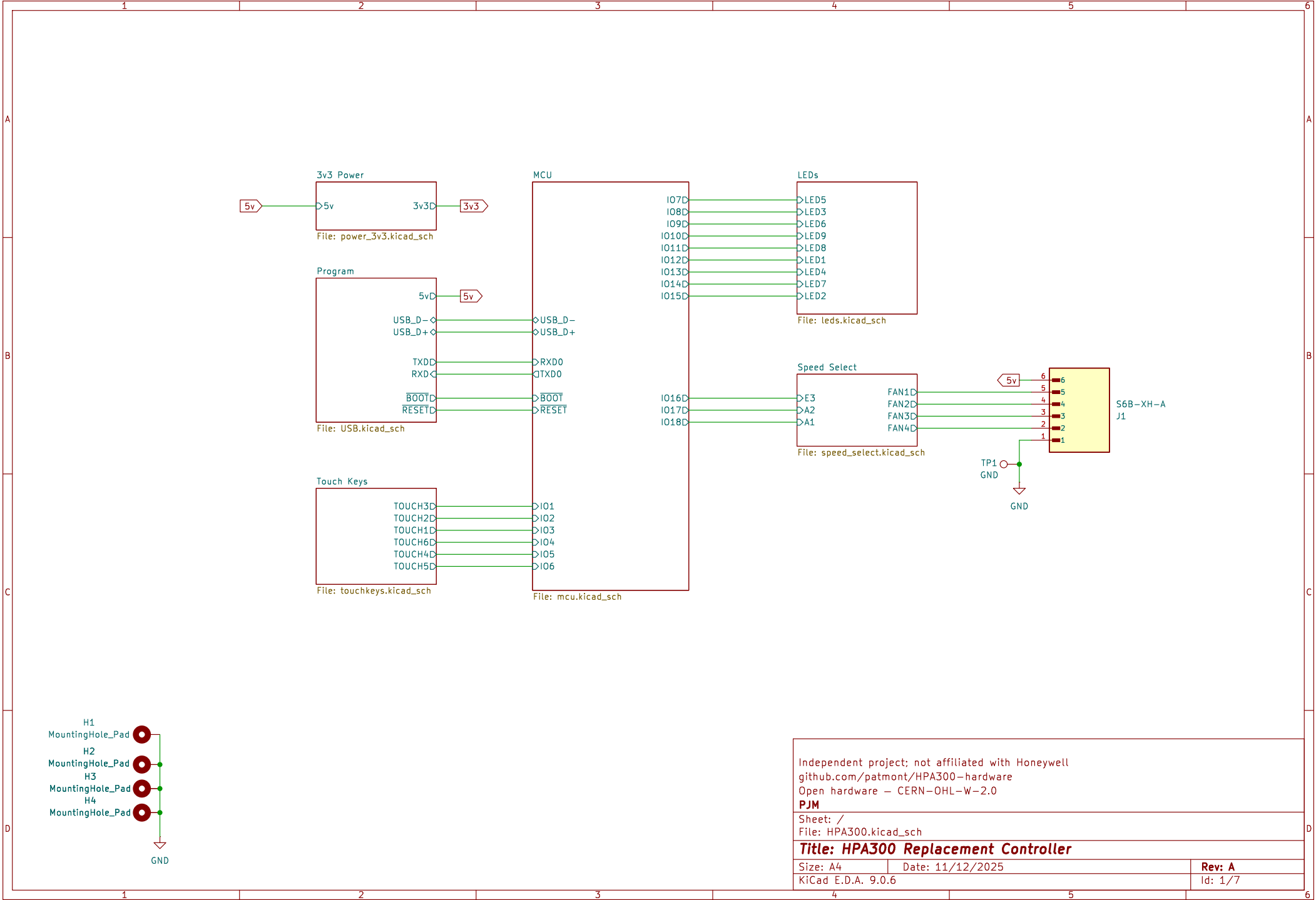
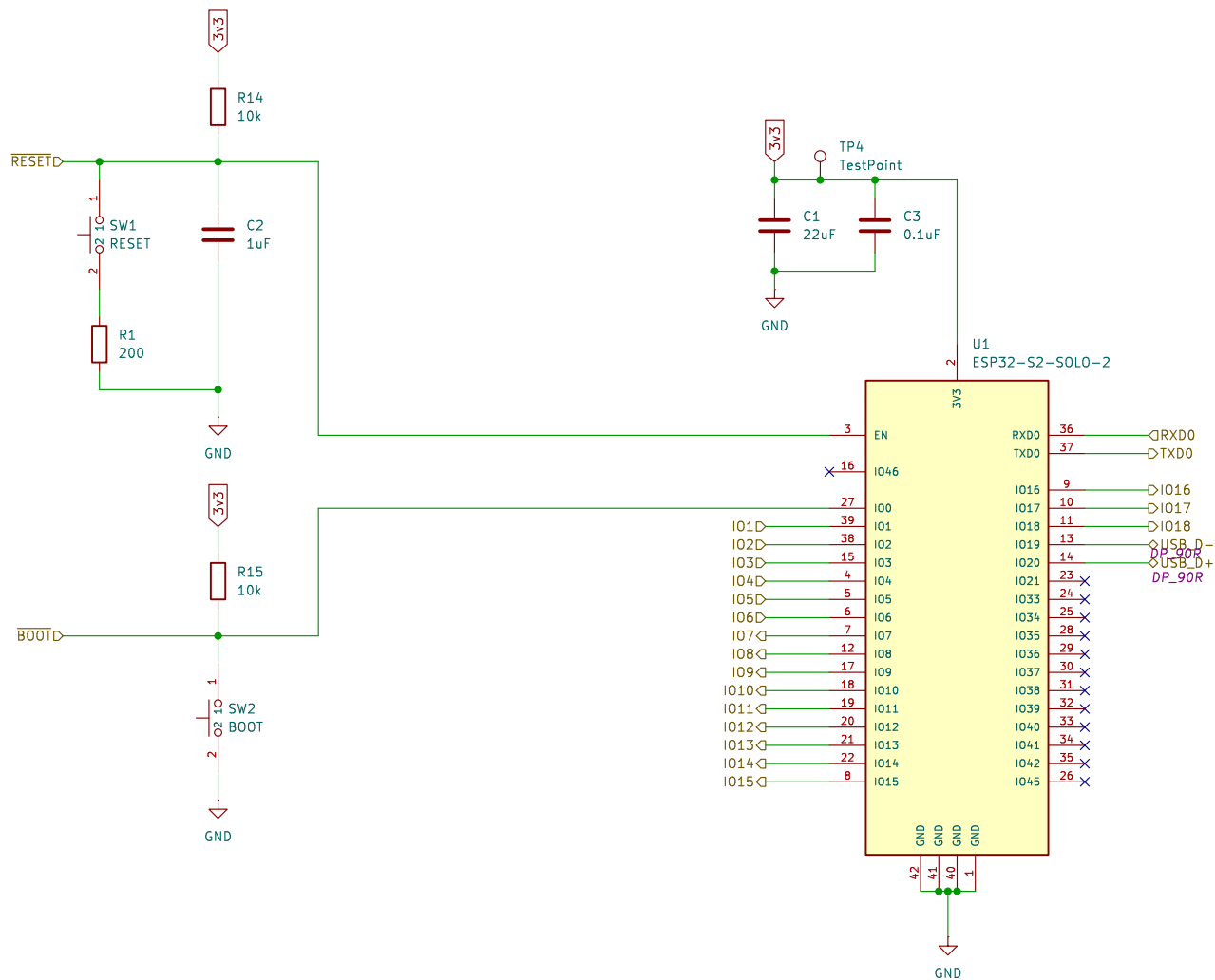






Name	No.	Type	Function
GND	1	P	Ground
3V3	2	P	Power supply
EN	3	I	High: on, enables the chip. Low: off, the chip powers off. Note: Do not leave the EN pin floating.
IO4	4	I/O/T	RTC_GPIO4, GPIO4, TOUCH4, ADC1_CH3
IO5	5	I/O/T	RTC_GPIO5, GPIO5, TOUCH5, ADC1_CH4
IO6	6	I/O/T	RTC_GPIO6, GPIO6, TOUCH6, ADC1_CH5
IO7	7	I/O/T	RTC_GPIO7, GPIO7, TOUCH7, ADC1_CH6
IO15	8	I/O/T	RTC_GPIO15, GPIO15, UORTS, ADC2_CH4, XTAL_32K_P
IO16	9	I/O/T	RTC_GPIO16, GPIO16, UORTS, ADC2_CH5, XTAL_32K_N
IO17	10	I/O/T	RTC_GPIO17, GPIO17, UORTS, ADC2_CH6, DAC_1
IO18	11	I/O/T	RTC_GPIO18, GPIO18, UORTS, ADC2_CH7, DAC_2, CLK_OUT3
IO8	12	I/O/T	RTC_GPIO8, GPIO8, TOUCH8, ADC1_CH7
IO19	13	I/O/T	RTC_GPIO19, GPIO19, UORTS, ADC2_CH8, CLK_OUT2, USB_D-
IO20	14	I/O/T	RTC_GPIO20, GPIO20, UORTS, ADC2_CH9, CLK_OUT1, USB_D+
IO3	15	I/O/T	RTC_GPIO3, GPIO3, TOUCH3, ADC1_CH2
IO46	16	I	GPIO46
IO9	17	I/O/T	RTC_GPIO9, GPIO9, TOUCH9, ADC1_CH8, FSPHID
IO10	18	I/O/T	RTC_GPIO10, GPIO10, TOUCH10, ADC1_CH9, FSPICSO, FSPHID4
IO11	19	I/O/T	RTC_GPIO11, GPIO11, TOUCH11, ADC2_CH0, FSPHID, FSPHID5
IO12	20	I/O/T	RTC_GPIO12, GPIO12, TOUCH12, ADC2_CH1, FSPICLK, FSPHID6
IO13	21	I/O/T	RTC_GPIO13, GPIO13, TOUCH13, ADC2_CH2, FSPHID, FSPHID7
IO14	22	I/O/T	RTC_GPIO14, GPIO14, TOUCH14, ADC2_CH3, FSPHID, FSPHID8
IO21	23	I/O/T	RTC_GPIO21, GPIO21
IO33	24	I/O/T	SPI04, GPIO33, FSPHID
IO34	25	I/O/T	SPI05, GPIO34, FSPICSO
IO45	26	I/O/T	GPIO45
IO0	27	I/O/T	RTC_GPIO0, GPIO0
IO35	28	I/O/T	SPI06, GPIO35, FSPHID
IO36	29	I/O/T	SPI07, GPIO36, FSPICLK
IO37	30	I/O/T	SPI08, GPIO37, FSPHID
IO38	31	I/O/T	GPIO38, FSPHID
IO39	32	I/O/T	MTCK, GPIO39, CLK_OUT3
IO40	33	I/O/T	MTDO, GPIO40, CLK_OUT2
IO41	34	I/O/T	MTDI, GPIO41, CLK_OUT1
IO42	35	I/O/T	MTMS, GPIO42
RXD0	36	I/O/T	UORTX, GPIO44, CLK_OUT2
TXD0	37	I/O/T	UORTX, GPIO43, CLK_OUT1
IO2	38	I/O/T	RTC_GPIO2, GPIO2, TOUCH2, ADC1_CH1
IO1	39	I/O/T	RTC_GPIO1, GPIO1, TOUCH1, ADC1_CH0

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github.com/patmont/HPA300-hardware
 Open hardware — CERN-OHL-W-2.0

PJM

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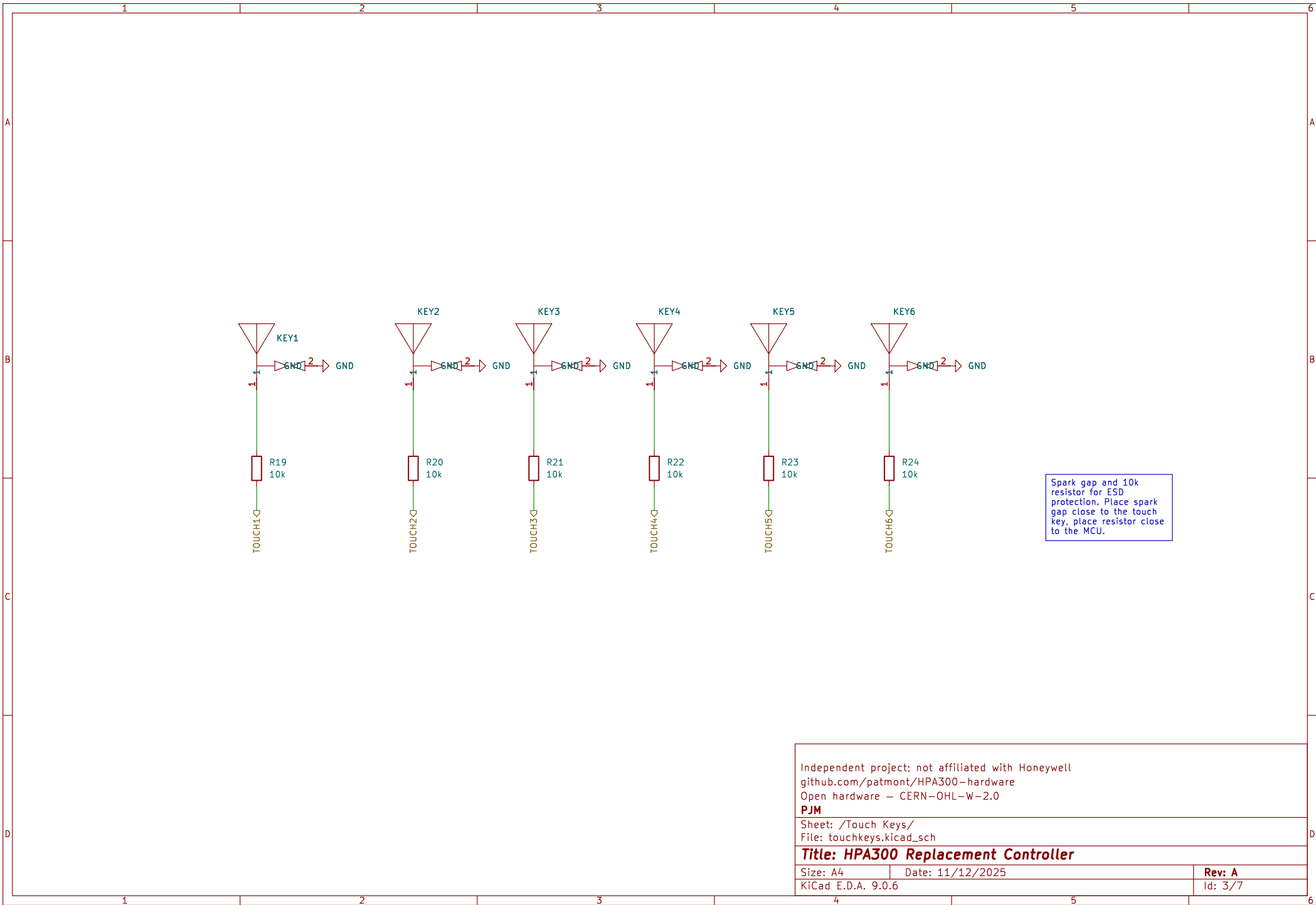
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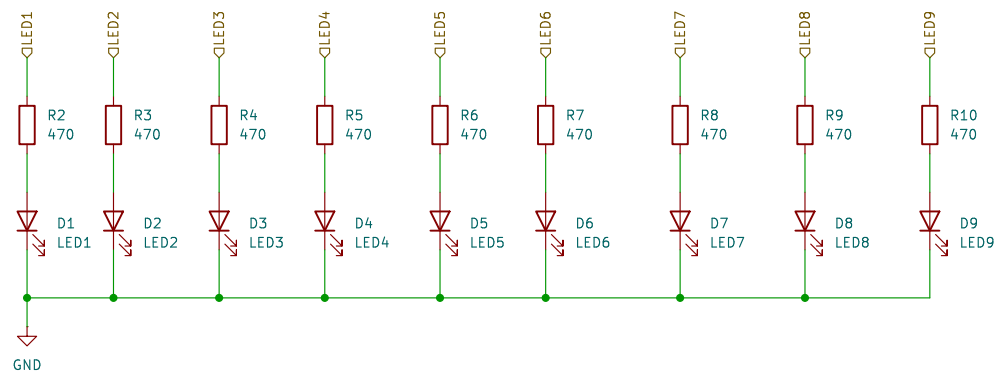
Size: A4 Date: 11/12/2025

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Rev: A

Id: 2/7





LED1: FAN 3
LED2: CHECK FILTERS
LED3: 8h TIMER
LED4: FAN 2
LED5: CHECK PRE-FILTER
LED6: 4h TIMER
LED7: FAN 1
LED8: FAN 4
LED9: 2h TIMER

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PJM

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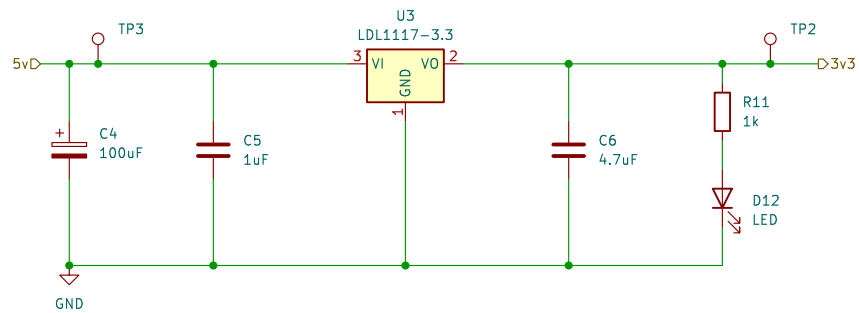
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Rev: A

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PJM

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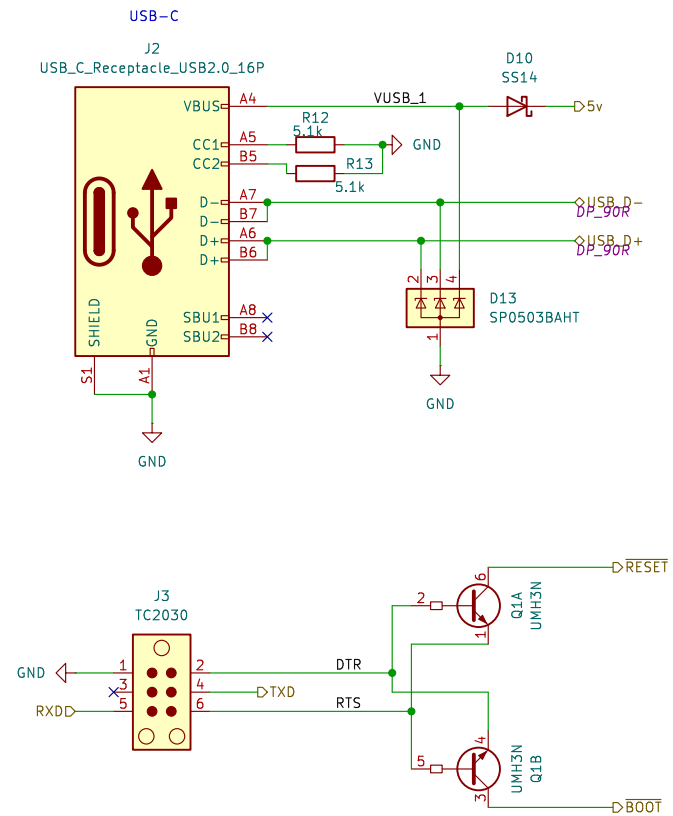
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Date: 11/12/2025

Rev: A

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Id: 5/7



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Title: HPA300 Replacement Controller

Size: A4 Date: 11/12/2025

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Rev: A

Id: 6/7

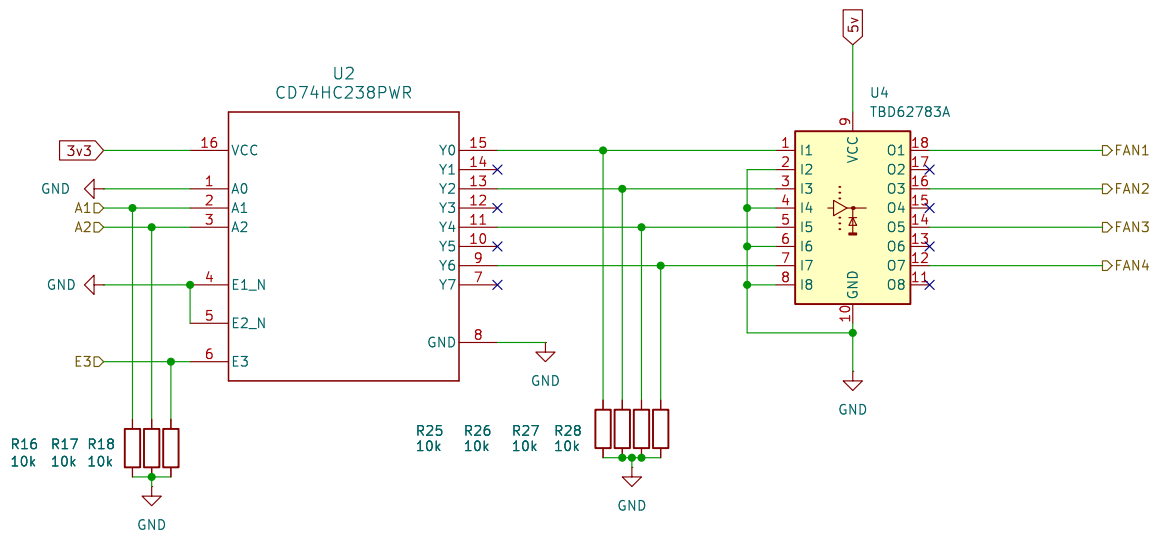


Table 7-2. Function Table 'HC238, 'HCT238

INPUTS			OUTPUTS										
STROBE			ADDRESS										
G2	G1	G0	A ₂	A ₁	A ₀	Y ₀	Y ₁	Y ₂	Y ₃	Y ₄	Y ₅	Y ₆	Y ₇
X	X	H	X	X	X	L	L	L	L	L	L	L	L
L	X	X	X	X	X	L	L	L	L	L	L	L	L
X	H	X	X	X	X	L	L	L	L	L	L	L	L
H	L	L	L	L	L	H	L	L	L	L	L	L	L
H	L	L	L	L	H	L	H	L	L	L	L	L	L
H	L	L	L	H	L	L	L	H	L	L	L	L	L
H	L	L	L	H	H	L	L	L	H	L	L	L	L
H	L	L	H	L	L	L	L	L	L	H	L	L	L
H	L	L	H	L	H	L	L	L	L	L	H	L	L
H	L	L	H	H	L	L	L	L	L	L	L	H	L
H	L	L	H	H	H	L	L	L	L	L	L	L	H

H = High Voltage Level, L = Low Voltage Level, X = Don't Care

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 Open hardware — CERN-OHL-W-2.0

PJM

Sheet: /Speed Select/
 File: speed_select.kicad_sch

Title: HPA300 Replacement Controller

Size: A4

Date: 11/12/2025

Rev: A

KiCad E.D.A. 9.0.6

Id: 7/7